

Philip-Chandan Brain Imaging Pipeline

UCSF longitudinal glioma: Phase 0 spike + 7-patient cohort export, expert segmentation, PDE prep

QBIHack brain-cancer-sim | Generated 2026-06-28

Philip-Chandan owns the brain imaging pipeline: download real glioma MRI with expert segmentations (not Otsu heuristics), validate NiftI pairs, export raw 3D numpy arrays, run prepare_pde_input.py (resample/crop/normalize to 64^3), and hand solver-ready cubes to Vinesh for PDE growth simulation and calibration. Phase 0 locked UCSF patient 100002; export_all_raw.py now batch-exports seven UCSF cohort patients (baseline + follow-up) with checkpointed progress and WT volume QC.

1. Phase 0 spike patient

Field	Value
Dataset	UCSF Longitudinal Glioma
Patient ID	100002
Diagnosis	Oligodendroglioma, WHO grade 2
IDH / MGMT	mut / positive
Timepoint	baseline (time1)
Slug	glioma_ucsf_100002_baseline

Unlike breast-cancer-sim (TCIA DICOM + Otsu), brain datasets ship NiftI volumes with radiologist-drawn tumor masks. We use those masks as ground truth for QC and for Vinesh's PDE initial condition per handoff_contract.json v1.0.0.

2. Pipeline modules (Phase 0)

Module	Role
nifti_extractor.py	Load NiftI as (Z,Y,X) float32; resolve UCSF paths; validate MR+mask pairs
export_raw_extract.py	Write raw .npy + JSON sidecar; copy expert mask to segmentations/
export_all_raw.py	Checkpointed cohort batch export + optional UCSF workbook volume QC
wt_volume_report.py	Longitudinal WT mm^3 from expert seg; compare to UCSF Table S1
pde_burden_compare.py	PDE 64^3 voxel burden vs WT growth % spec
qc_slice_plot.py	Mid-Z PNG + longitudinal before/after overlay panels
view_volume_napari.py	Interactive 3D QC viewer with clinical display controls
cohort/cohort_discovery.py	Scan local NiftI trees; find longitudinal cases; audit cohort.json
download_mu_glioma_post.py	TCIA Faspex helper for MU-Glioma-Post backup dataset
../vinesh/prepare_pde_input.py	Resample expert mask, normalize MR, crop to 64^3 PDE input
qc_pde_plot.py	QC PNGs for resampled expert mask + cropped PDE volume slices

On-disk outputs

Artifact	Path
Raw MR volume	data/processed/raw-extract-philip-chandan/100002/baseline.npy
Metadata sidecar	data/processed/raw-extract-philip-chandan/100002/baseline.json
Expert mask	data/processed/segmentations/glioma_ucsf_100002_baseline_mask.nii.gz
QC overlay PNG	data/qc/slice-plots-philip-chandan/glioma_ucsf_100002_baseline_mid-z-overlay.png
Longitudinal QC	data/qc/slice-plots-philip-chandan/{patient_id}_longitudinal_mid-z-overlay.png
PDE prep QC	data/qc/pde-prep-vinesh/g64/glioma_ucsf_100002_baseline_pde-input-mid-z.png
PDE input	data/processed/pde-input-vinesh/100002/g64/baseline.npy

Spike volume metadata

Property	Value
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Shape (Z,Y,X)	[155, 240, 240]
Spacing (mm)	[1.0, 1.0, 1.0]
Source MR	data/raw/ucsf_alptdg/100002/100002_time1_t1ce.nii.gz
Tumor voxels	8,137 (expert mask, full volume)

PDE input metadata

Property	Value
Shape (Z,Y,X)	[64, 64, 64]
Spacing (mm)	[1.0, 1.0, 1.0]
Segmentation	expert_mask
Tumor voxels	8,137 (cropped 64^3 grid)
Source mask	data/processed/segmentations/glioma_ucsf_100002_baseline_mask.nii.gz

3. Visual QC - expert segmentation overlay

qc_slice_plot.py picks the axial slice with the most tumor voxels and draws a lime contour from the UCSF expert mask. This confirms mask alignment before export and before handing data to Vinesh.

glioma_ucsf_100002_baseline z=81

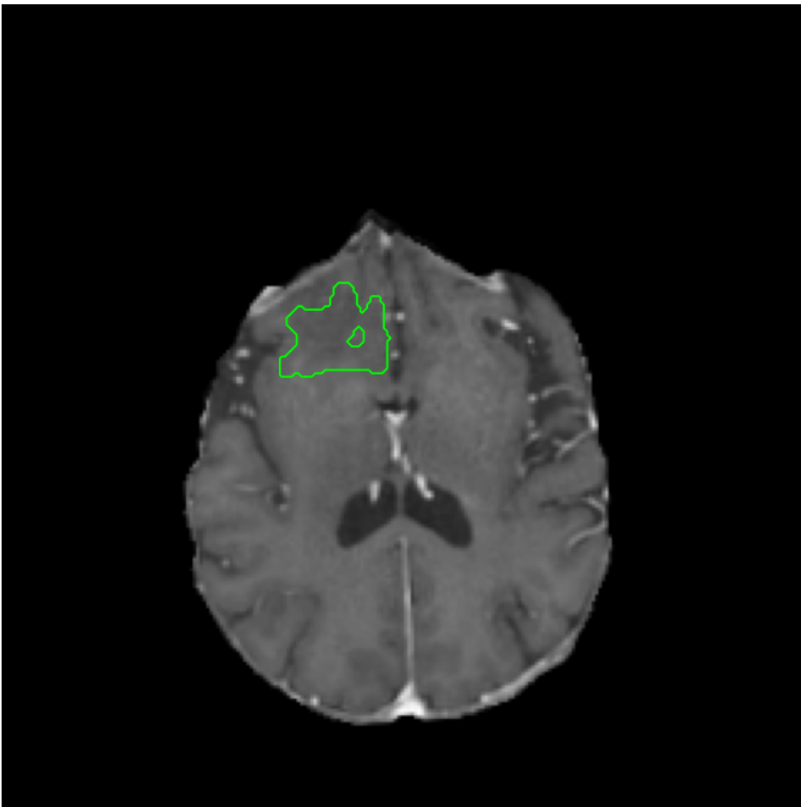


Figure 1. UCSF patient 100002 baseline T1ce with expert segmentation contour (lime). Slice chosen at maximum tumor cross-section.

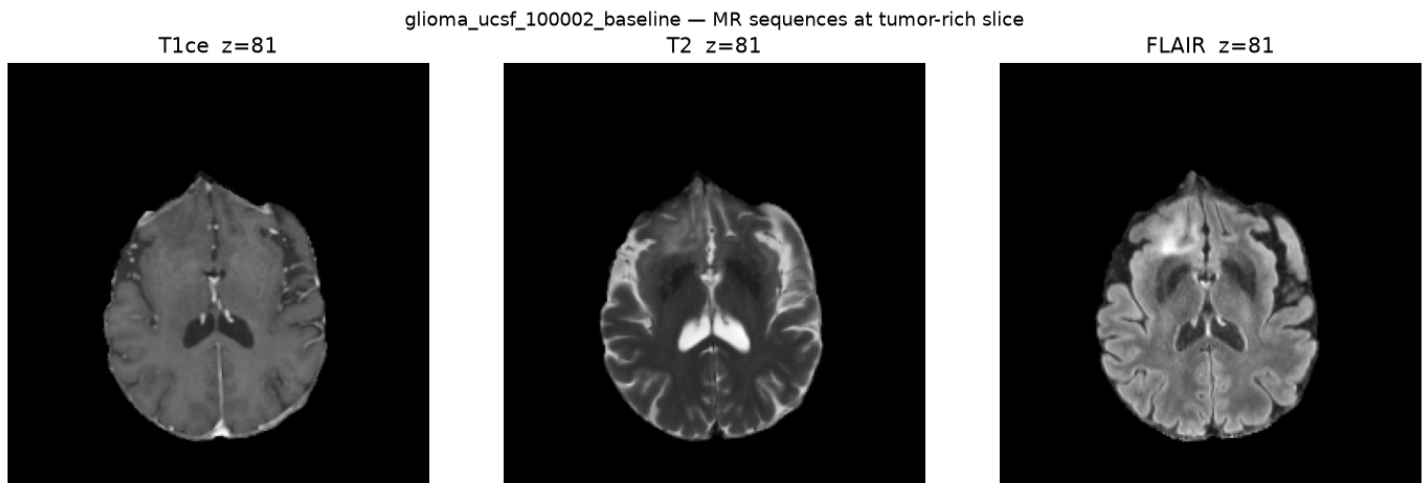


Figure 2. Same z slice across MR sequences available for this timepoint. Low-grade gliomas may show subtle T1ce enhancement; FLAIR often highlights edema.

4. MRI sequences - what T1, T2, FLAIR, and T1ce mean

Hospital brain MRI protocols acquire several contrasts of the same anatomy. Radiologists read them together; our napari viewer loads all UCSF series for the spike timepoint so you can toggle layers like on a PACS workstation.

Sequence	Full name	What you see	Glioma relevance
T1	T1-weighted	Anatomy; CSF dark, fat bright	Baseline anatomy; pre-contrast reference
T1ce	T1 post-contrast (T1+C)	Areas with leaky blood vessels brighten after gadolinium	Enhancing tumor rim; spike export uses T1ce as primary MR
T2	T2-weighted	Fluid bright; good soft-tissue contrast	Tumor and edema often appear brighter than normal brain
FLAIR	Fluid-attenuated inversion recovery	T2-like but CSF suppressed (dark ventricles)	Peritumoral edema and infiltrative signal stand out clearly

For our grade 2 IDH-mut oligodendroglioma spike, T1ce enhancement is subtle - mean intensity inside the expert mask is almost identical to surrounding brain. FLAIR is often more informative for edema. The expert mask delineates the lesion using all available clinical context, not brightness on a single sequence.

5. Clinical napari viewer - view_volume_napari.py

We extended the napari QC viewer beyond a simple grayscale overlay. Controls live in the right dock panels. CLAHE and window/level adjustments are labeled QC-only - they help engineers inspect data, not replace diagnostic PACS reading.

Window / Level (WW/WL)

Radiology displays map MR intensities to gray using a window (width) and level (center). Narrow windows increase contrast but clip bright/dark regions. Our defaults compute level and width from brain voxels only (2nd-98th percentile), ignoring black padding outside the head. Sliders update all MR layers; hold Ctrl and drag on the canvas for click-drag WW/WL like a clinical viewer.

Multi-sequence layers

For UCSF patient folders, the viewer auto-loads T1, T1ce, T2, and FLAIR when NiftI files exist. T1ce is visible by default with the expert overlay. Toggle FLAIR or T2 in napari's layer list to compare edema vs enhancement on the same slice.

Plane selector and orthogonal MPR grid

The Plane dropdown switches axial, coronal, or sagittal viewing on the main canvas while keeping the segmentation aligned. Orthogonal MPR grid opens a 1x3 linked layout (axial | coronal | sagittal) for navigating the tumor in three planes; the overlay is hidden in grid mode because napari assigns one layer per grid cell.

CLAHE (Contrast Limited Adaptive Histogram Equalization)

CLAHE enhances local contrast slice-by-slice inside the brain mask. It can make subtle texture differences easier to see during QC. It is not a physical imaging sequence and must not be used for diagnosis or as PDE input - it is a display filter only, clearly marked in the viewer.

Overlay toggle

Hide overlay / Show overlay toggles the expert segmentation labels so you can inspect underlying MR intensity. For subtle T1ce lesions, pairing this with FLAIR and WW/WL adjustment is the intended workflow.

Launch command

```
cd brain-cancer-sim
.venv/bin/python simulation-vinesh-philip-chandan/philip-chandan/view_volume_napari.py --slug glioma_ucsf_100002_baseline
```

6. Expert-mask PDE prep - prepare_pde_input.py (Philip-Chandan)

Philip-Chandan runs `prepare_pde_input.py` after each raw export, writing PDE-ready volumes to `data/processed/pde-input-vinesh/<patient_id>/g64/` per `handoff_contract.json v1.0.0`. Unlike `breast-cancer-sim` (Otsu on DCE-MRI), `brain v1` uses the dataset expert mask - no Otsu fallback. Vinesh consumes these cubes for `run_growth()` and `calibrate.py`.

- Resample MR (linear) and expert mask (nearest) to isotropic 1 mm spacing
- Min-max normalize MR intensities to [0, 1]
- Keep continuous normalized density inside the expert mask; zero background
- Crop/pad to max 64^3 centered on mask center of mass

Continuous tumor values (not a binary mask) are required so the PDE logistic growth term $\rho \cdot u \cdot (1-u)$ can evolve. `qc_pde_plot.py` documents the same pipeline via `prepare_pde_stages()` for the PDF figures below.

100002 g64 expert mask z=81 (mask vox=8137)

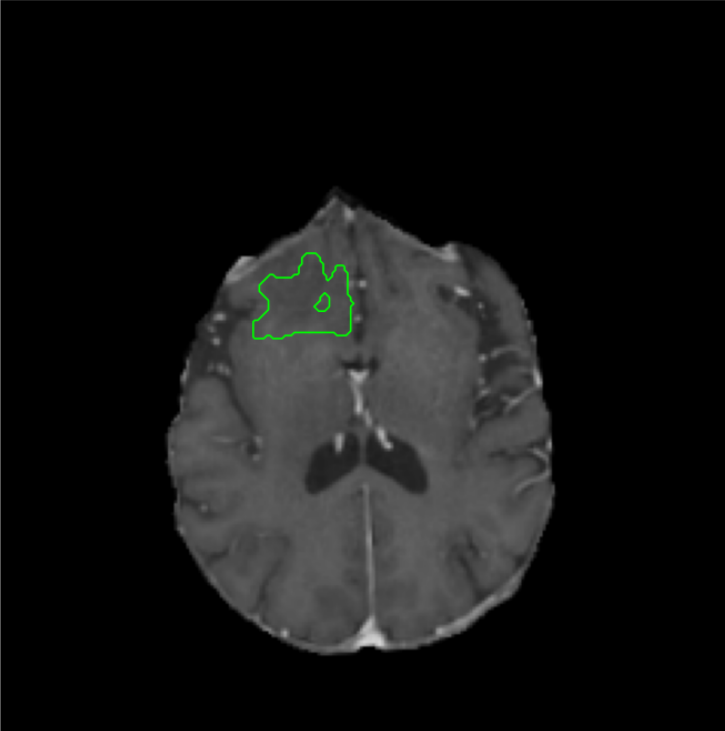


Figure 3. UCSF patient 100002 after resample + normalize. Lime contour is the expert tumor mask on the 1 mm grid before crop; background will be zeroed.

Patient 100002 g64 cropped PDE input (inferno, cyan contour > 0)
 Baseline (t1) z=30 shape=[64, 64, 64] Follow-up (t2) z=29 shape=[64, 64, 64]

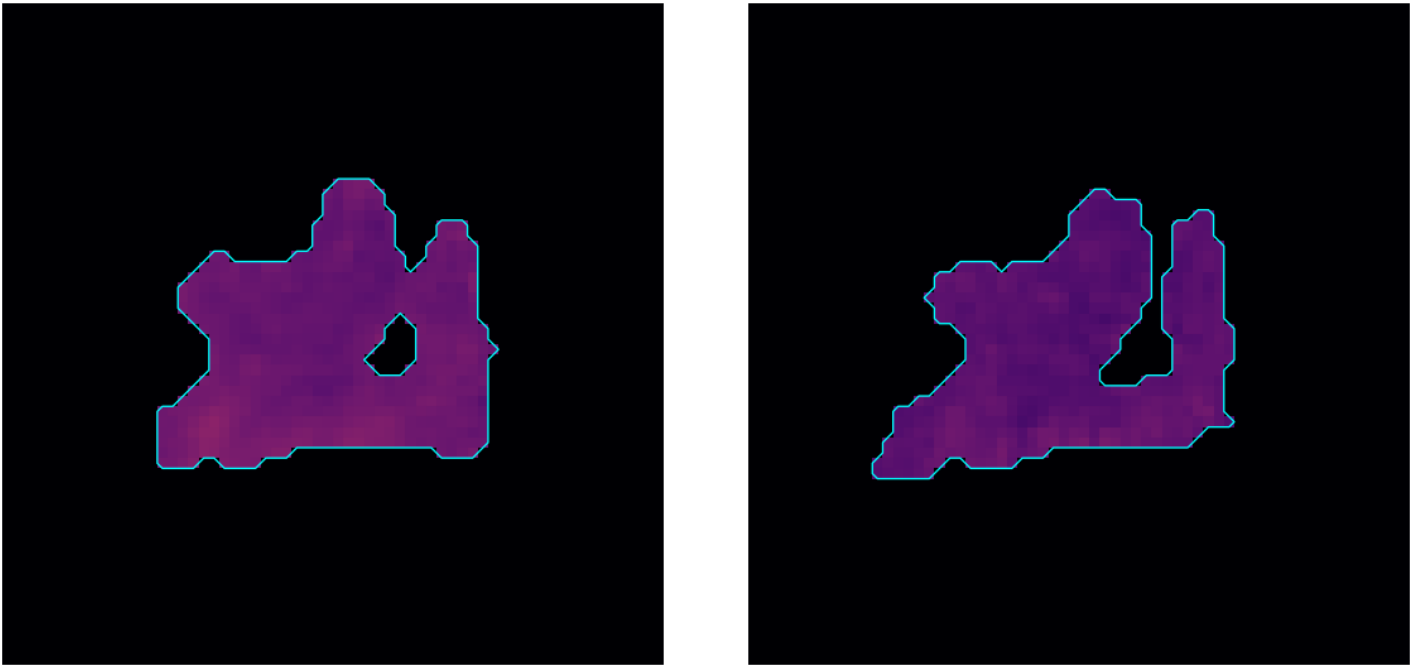


Figure 4. Cropped PDE input (64^3 , 1 mm) for spike patient 100002. Left: baseline (t1); right: follow-up (t2). Inferno colormap shows continuous initial tumor burden; cyan contour marks voxels > 0 .

Solver smoke test on spike PDE input

run_growth() on glioma_ucsf_100002_baseline: 51 frames, shape [64, 64, 64], tumor volume ($u > 0$) at $t=0$ = 262144 mm^3 , at $t=\text{end}$ ($u \geq 0.5$) = 5738 mm^3 . Growth term is active because initial burden is continuous inside the expert ROI.

Regenerate PDE artifacts

```
cd brain-cancer-sim
.venv/bin/python simulation-vinesh-philip-chandan/vinesh/prepare_pde_input.py
.venv/bin/python simulation-vinesh-philip-chandan/philip-chandan/qc_pde_plot.py
```

7. PDE burden vs WT growth spec (QC)

After prepare_pde_input.py, tumor burden in the 64^3 PDE cube is the count of voxels with value > 0 (handoff_contract tumor_burden_rule). At 1 mm spacing each voxel is $\sim 1 \text{ mm}^3$ when the lesion fits inside the crop. WT volumes in wt_volume_report.json use BraTS labels 1+2+3 only; PDE prep seeds from mask > 0 , so per-timepoint capture can exceed 100% when label 4 (resection cavity) is present. The longitudinal check is whether PDE voxel growth % tracks computed_growth_pct from the expert WT spec.

Per-timepoint burden capture (WT mm^3 vs PDE voxels)

Patient	TP	WT mm^3	PDE voxels	Capture
100002	t1	8,137	8,137	100.0%
100002	t2	8,365	8,365	100.0%
100118	t1	10,905	10,881	99.8%
100118	t2	77,327	72,755	94.1%
100192	t1	11,463	16,941	147.8%
100192	t2	4,798	10,028	209.0%
100260	t1	18,547	33,600	181.2%

100260	t2	97,533	80,262	82.3%
100220	t1	38,492	38,842	100.9%
100220	t2	108,441	92,327	85.1%
100134	t1	11,235	65,801	585.7%
100134	t2	12,288	37,584	305.9%
100130	t1	4,144	11,836	285.6%
100130	t2	4,722	8,313	176.0%

Longitudinal growth % - WT spec vs PDE voxels

Patient	Cap t1	Cap t2	WT d%	PDE d%	d diff	QC flags
100002	100.0%	100.0%	+2.8%	+2.8%	+0.0	ok
100118	99.8%	94.1%	+609.1%	+568.6%	-40.5	crop_loss_followup, growth_mismatch
100192	147.8%	209.0%	-58.1%	-40.8%	+17.3	label_inflation_baseline, label_inflation_followup, growth_mismatch
100260	181.2%	82.3%	+425.9%	+138.9%	-287.0	label_inflation_baseline, crop_loss_followup, growth_mismatch
100220	100.9%	85.1%	+181.7%	+137.7%	-44.0	crop_loss_followup, growth_mismatch
100134	585.7%	305.9%	+9.4%	-42.9%	-52.3	label_inflation_baseline, label_inflation_followup, growth_mismatch
100130	285.6%	176.0%	+13.9%	-29.8%	-43.7	label_inflation_baseline, label_inflation_followup, growth_mismatch

Spike patient 100002 matches exactly (100% capture, +2.8% growth). 0/7 patients pass all QC flags (capture 95-105% both timepoints and growth within +/-5% of WT spec). Large-tumor patients (e.g. 100118, 100220, 100260) show follow-up crop loss that skews PDE growth % - Vinesh calibrate.py should use full-res WT targets, not cropped voxel counts alone. JSON sidecar: data/processed/pde-input-vinesh/pde_burden_compare.json.

8. UCSF cohort scale-up - 7 patients longitudinal QC

export_all_raw.py exported baseline + follow-up raw extracts for seven UCSF-LPTDG patients in cohort.json (primary spike 100002 plus six discovery picks). qc_slice_plot.save_longitudinal_overlay_plot() renders side-by-side T1ce slices at the baseline tumor-rich z index with lime expert contours (PNG on disk under data/qc/slice-plots-philip-chandan/). WT volumes (labels 1+2+3) come from wt_volume_report.py; use --compare-ucsf-workbook on export_all_raw to diff against UCSF Table S1.

Patient	Grade	IDH	WT t1 (mm3)	WT t2 (mm3)	d mm3	d %	Days
100002	2	mut	8,137	8,365	+228	+2.8%	183
100118	4.0	WT	10,905	77,327	+66,422	+609.1%	56
100192	2.0	mut	11,463	4,798	-6,665	-58.1%	83
100260	4.0	WT	18,547	97,533	+78,986	+425.9%	63
100220	4.0	WT	38,492	108,441	+69,949	+181.7%	27
100134	3.0	mut	11,235	12,288	+1,053	+9.4%	105
100130	3.0	mut	4,144	4,722	+578	+13.9%	182

9. Handoff contract summary

Philip-Chandan (raw extract)	Vinesh (PDE input)
Raw MR intensities, not normalized	Philip-Chandan: prepare_pde_input -> 64^3, 1 mm
spacing_mm in JSON sidecar	Normalize to [0, 1] inside tumor region
Expert mask path in metadata	Expert mask resampled - no Otsu fallback
Axis order (Z, Y, X)	Vinesh: run_growth() + calibrate.py

10. Summary and next steps

Phase 0 delivered NIfTI ingest, UCSF spike export, expert-mask QC, napari viewer, Philip-Chandan PDE prep into 64^3 continuous-density input, and solve_growth() smoke on patient 100002. Cohort scale-up exported seven patients (14 timepoints) with longitudinal overlay PNGs on disk, WT volume table (section 8), and PDE burden vs growth QC (section 7). Next: manifest.json v1.0.0, Vinesh calibrate.py for t1->t2, and demo toggle (100002 stable IDH-mut vs 100118 aggressive IDH-WT GBM).

Deliverables completed

Stage	Deliverable	Status
Cohort	cohort.json rev1 - 7 UCSF patients	Done
Export	export_all_raw.py checkpointed batch	7 patients × 2 timepoints
Volume QC	wt_volume_report.json + workbook compare	Done
Longitudinal QC	7 before/after overlay PNGs	On disk
PDE prep	prepare_pde_input g64 - 7 patients x 2 timepoints	Done
PDE burden QC	pde_burden_compare.json + section 7	In this PDF
Solver	run_growth() smoke on 100002 baseline	Passed
Pending	manifest.json, calibrate.py, Jasim/Vihari	Next